Two analytic theorems for log-cooled Gaussian attention: planted partitions at the $\sqrt{c}$ scale, and nonlocal rescue in three-layer chains

Theory notes for the ICML 2027 draft (v2/theory)

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Abstract

We replace two weak points of the current draft by analytic theorems. **(A)** For an arbitrary graph with a planted partition and vector features, the planted partition is the limit of the log-cooled Gaussian-attention dynamics whenever every block has feature diameter at most $\sqrt{c}$ (up to a drift budget 2η) and adjacent blocks are separated by $\Gamma(t_0)\sqrt{c}$, where $\Gamma(t_0) = 1 + (1 + o(1)) \log\log t_0 / \log t_0$ is asymptotically the sharp two-block threshold $\kappa(t_0)$. In ball form: within-block radius $r \le \sqrt{c}\,(1/2 - \eta)$ and mean separation $\Delta \ge 2r + \Gamma(t_0)\sqrt{c}$. This removes the factor $1/(4\sqrt{\rho})$ (hop radius ρ) of the old condition. The mixing constant is sharp among conditions that depend only on block diameters. On the E4 data, the new condition certifies every run with $\sigma \le 0.1$ (100/100), where the old one certified only $\sigma \le 0.03$. **(B)** For the three-layer chain we prove an exact persistence criterion: gap 1 persists iff $u + v/2$ never drops below 1. We prove a one-step rescue theorem valid for *every* $t_0 \ge 2$ and *every* $v_0 > 0$, with explicit depth. Finally we determine the $t_0 \to \infty$ scaling limit of the normalised rescue boundary: $u^*(v_0)/\kappa(t_0) \to \varphi(v_0/\kappa(t_0))$ with $\varphi(x) = \max\{x, 1 - x/2\}$ for $x < 1$ and $\varphi = 1$ for $x > 1$. Its minimum is $2/3$, at $x = 2/3$, which explains the observed drift $0.770 \to 0.682$.

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1 Setting, standing facts, and a universal dichotomy

Model and units. $G = (V, E)$ is a finite undirected graph with a self-loop at every node. $N[i]$ is the closed neighbourhood of i and $N_{\max} := \max_i |N[i]|$. Features $z_i(t) \in \mathbb{R}^p$ follow

$$z_i(t+1) = \sum_{j \in N[i]} \alpha_{ij}(t)\, z_j(t), \qquad \alpha_{ij}(t) = \frac{w_{ij}(t)}{\pi_i(t)}, \quad w_{ij}(t) = s_t^{-\|z_i(t) - z_j(t)\|^2}, \quad \pi_i(t) = \sum_{j \in N[i]} w_{ij}(t), \tag{1}$$

with $s_t := t + t_0$, $t_0 > 1$, and $L_t := \log s_t$. All lengths are in units of $\sqrt{c}$, i.e. the schedule $T_t = c/\log(t + t_0)$ has been absorbed (a fixed metric W is absorbed by $z = Wh$, as in the main text). Note $w_{ii} = 1$ and $0 < w_{ij} \le 1$, hence

$$1 \le \pi_i(t) \le N_{\max}. \tag{2}$$

An edge $\{i, j\}$ *merges* if $\|z_i(t) - z_j(t)\| \to 0$ and *persists* if $\liminf_t \|z_i(t) - z_j(t)\| > 0$. For chains these words refer to the gaps.

1.1 The isolated two-layer recursion

Let $F_t(d) := d\,\frac{1-a}{1+a}$ with $a = s_t^{-d^2}$, and let $d_{t+1} = F_t(d_t)$, $d_0 > 0$ (offset t_0). The next lemma collects the parts of Theorem 1 of the main text that are used below, with short proofs so that this note is self-contained. The sharp asymptotics $\kappa(t_0) = 1 + (1 + o(1)) \log\log t_0 / \log t_0$ (Theorem 1(v) of the main text) is quoted only in remarks and is never used in a proof.

Lemma 1.1 (two-layer facts)**.** *Let $t_0 > 1$.*

(a) F_t is a continuous increasing bijection of $(0, \infty)$, $F_t(d) \le d$, and for fixed t, d the value $F_t(d)$ is nondecreasing in t_0.

(b) If $d_T \le 1$ for some T then $d_t \to 0$; otherwise $d_t > 1$ for all t and $\lim_t d_t \ge 1$.

(c) The merge set $M(t_0) = \{d_0 :\ d_t \to 0\}$ equals $(0, \kappa(t_0))$ for some $\kappa(t_0) \in (1, \infty)$, and $d_0 = \kappa(t_0)$ persists.

(d) (critical curve) $\kappa(t_0 + k) = F_{k-1} \circ \cdots \circ F_0(\kappa(t_0)) \le \kappa(t_0)$ for $k \in \mathbb{N}$.

(e) κ is nonincreasing on $(1, \infty)$.

(f) (certificate C1) If $1 < \delta < d_0$ and, with $q := t_0^{-\delta^2}$,

$$\log d_0 - \Big(2 + \frac{q}{1 - q^2}\Big) \frac{(t_0 - 1)^{1-\delta^2}}{\delta^2 - 1} \ \ge\ \log \delta, \tag{3}$$

then $d_t \ge \delta$ for all t. Hence $\kappa(t_0) \le \kappa_+(t_0) := \inf\{d_0 :\ (3) \text{ holds for some } \delta\}$. The function κ_+ is nonincreasing and $\kappa_+(t_0) \to 1$ as $t_0 \to \infty$; in particular $\kappa(t_0) \downarrow 1$.

Proof. (a) $F_t'(d) = \frac{1-a}{1+a} + \frac{4d^2 L_t a}{(1+a)^2} > 0$, $F_t(0^+) = 0$, $F_t(\infty) = \infty$. For fixed d, a decreases as s_t grows, and $\frac{1-a}{1+a}$ increases.

(b) If $d_T \le 1$ then $d_t \le 1$ for $t \ge T$ by (a), so $a_t \ge s_t^{-1}$ and $d_{t+1} \le (1 - a_t) d_t \le e^{-a_t} d_t$. Since $\sum_t s_t^{-1} = \infty$, $d_t \to 0$. Otherwise the nonincreasing sequence stays above 1.

(c) By (b), $M(t_0) = \bigcup_T \{d_0 : \ d_T(d_0) < 1\}$ (if $d_T = 1$ then $d_{T+1} < 1$). Each $d_0 \mapsto d_T(d_0)$ is continuous and increasing, so each set in the union is an interval $(0, c_T)$ and $M(t_0) = (0, \kappa)$ is open with $\kappa = \sup_T c_T$. Since $d_1(1) < 1$, we get $\kappa > 1$; part (f) gives $\kappa < \infty$. As $\kappa \notin M(t_0)$, it persists.

(d) The trajectory from d_0 at offset t_0, after k steps, continues as the trajectory at offset $t_0 + k$ started from $\Phi_k(d_0) := F_{k-1} \circ \cdots \circ F_0(d_0)$. Hence $M(t_0 + k) = \Phi_k(M(t_0)) = (0, \Phi_k(\kappa(t_0)))$, using that Φ_k is an increasing bijection.

(e) Let $t_0 \le t_0'$ and let d_t, d_t' be the trajectories from the same d_0 at offsets t_0, t_0'. By (a), $d_{t+1}' = F_t'(d_t') \ge F_t'(d_t) \ge F_t(d_t) = d_{t+1}$. So $M(t_0') \subseteq M(t_0)$.

(f) Suppose $d_u \ge \delta$ for all $u < t$. Then $a_u \le s_u^{-\delta^2} \le q$ and $-\log\frac{1-a}{1+a} = -\log(1-a^2) + 2\log(1+a) \le \frac{a^2}{1-a^2} + 2a \le a\big(2 + \frac{q}{1-q^2}\big)$. Using $\sum_{u\ge 0}(t_0+u)^{-\delta^2} \le \int_{t_0-1}^{\infty} x^{-\delta^2}dx = \frac{(t_0-1)^{1-\delta^2}}{\delta^2-1}$, we get $\log d_t \ge \log\delta$, and induction closes. The left side of (3) is nondecreasing in t_0, so κ_+ is nonincreasing. Given $\epsilon > 0$, the pair $d_0 = 1+2\epsilon$, $\delta = 1+\epsilon$ satisfies (3) for large t_0, since the middle term tends to 0. □

1.2 Chains: comparison

For the layered graph $G_{r,m}$ with layer-constant features (a property preserved by the dynamics), write $w_l = s_t^{-g_l^2}$, $w_0 = w_r = 0$, $D_l = 1 + w_{l-1} + w_l$, $A = 1 + w_{l-1}$, $B = 1 + w_{l+1}$. The gaps obey (main text, `core.py`)

$$g_l' = g_l \frac{AB - w_l^2}{D_l D_{l+1}} + \frac{w_{l+1} g_{l+1}}{D_{l+1}} + \frac{w_{l-1} g_{l-1}}{D_l}. \tag{4}$$

Lemma 1.2 (comparison; main text)**.** *All gaps stay nonnegative, and $g_l(t+1) \ge F_t(g_l(t))$ for every l and t.*

Proof. All terms of (4) are nonnegative. It remains to show $(AB - w^2)(1+w) \ge (1-w)(A+w)(B+w)$ for $A, B \ge 1$, $w \in [0,1]$. The difference equals $w\,[(A-1)B + A(B-1) + w(A+B-2)] \ge 0$. □

Lemma 1.3 (Theorem 2 of the main text)**.** *If $g_l(t_1) \ge \kappa(t_0 + t_1)$ for some t_1, then $g_l(t) \ge 1$ for all $t \ge t_1$; in particular the gap persists.*

Proof. By Lemma 1.2 and the monotonicity of F_t, induction gives $g_l(t_1 + k) \ge d_k$, where d is the isolated trajectory at offset $t_0 + t_1$ started from $\kappa(t_0 + t_1)$. By Lemma 1.1(b),(c), $d_k > 1$ for all k. □

1.3 Infinite flow and a universal dichotomy

We use the following theorem for products of stochastic matrices without a lower bound on the positive entries. (Hendrickx–Tsitsiklis [2], Thm. 2, needs such a bound in discrete time and therefore does *not* apply here, because attention weights decay to 0.)

Lemma 1.4 (Touri–Nedić [1], Thm. 1, deterministic case; see also their Thms. 4 and 7)**.** *Let $A(t)$, $t \ge 0$, be row-stochastic $n \times n$ matrices, and suppose there are $\alpha, \gamma > 0$ such that for all t: (i)* strong aperiodicity *$A_{ii}(t) \ge \gamma$ for all i; (ii)* balance *$\sum_{i\in S, j\notin S} A_{ij}(t) \ge \alpha \sum_{i\notin S, j\in S} A_{ij}(t)$ for every nonempty proper $S \subset [n]$. Then for every $x(0) \in \mathbb{R}^n$ the sequence $x(t) = A(t-1)\cdots A(0)x(0)$ converges. Moreover $\lim_t x_i(t) = \lim_t x_j(t)$ whenever i, j lie in the same connected component of the* infinite flow graph *$\{\{i,j\} : \ \sum_t (A_{ij}(t) + A_{ji}(t)) = \infty\}$.*

(Touri–Nedić state the theorem for independent random chains; a deterministic chain is a special case. Their Theorem 4 requires "weak aperiodicity", which follows from (i) by keeping two terms of the column inner product: $\sum_l A_{li}A_{lj} \ge A_{ii}A_{ij} + A_{jj}A_{ji} \ge \gamma(A_{ij} + A_{ji})$.)

Proposition 1.5 (limits exist; universal 0-or-$\sqrt{c}$ dichotomy)**.** *For every graph G with self-loops, every $t_0 > 1$ and every $z(0) \in \mathbb{R}^{n\times p}$:*

(i) every $z_i(t)$ converges to some z_i^∞;

(ii) if $\sum_t w_{ij}(t) = \infty$ for an edge $\{i, j\}$, then $z_i^\infty = z_j^\infty$;

(iii) for every edge, $\ell_{ij} := \|z_i^\infty - z_j^\infty\| \in \{0\} \cup [1, \infty)$.

Proof. Apply Lemma 1.4 to each coordinate with $A_{ij}(t) = \alpha_{ij}(t)$; these matrices depend on the trajectory, but for a fixed initial condition they form a fixed deterministic sequence. By (2), $A_{ii} = 1/\pi_i \geq 1/N_{\max}$. Since w is symmetric,

$$\sum_{i\in S, j\notin S} \frac{w_{ij}}{\pi_i} \;\geq\; \frac{1}{N_{\max}} \sum_{i\in S, j\notin S} w_{ij} = \frac{1}{N_{\max}} \sum_{i\notin S, j\in S} w_{ij} \;\geq\; \frac{1}{N_{\max}} \sum_{i\notin S, j\in S} \frac{w_{ij}}{\pi_i},$$

so (ii) of Lemma 1.4 holds with $\alpha = 1/N_{\max}$. Also $A_{ij} + A_{ji} \geq 2w_{ij}/N_{\max}$, which gives (i) and (ii). For (iii): if $\ell_{ij} \in (0, 1)$, pick $\ell' \in (\ell_{ij}, 1)$. Eventually $\|z_i - z_j\| < \ell'$, so $w_{ij}(t) \geq s_t^{-\ell'^2}$ and $\sum_t w_{ij} = \infty$, which contradicts (ii). □

Remark 1.6. Proposition 1.5(iii) extends the first-order transition of Theorem 1(i),(iii) of the main text (the jump of d_∞ from 0 to $\sqrt{c}$) to *every edge of every graph*, with vector features. Limiting edge lengths in $(0, \sqrt{c})$ are impossible.

2 Task A: planted partitions at the $\sqrt{c}$ scale

Setting. Let $V = B_1 \sqcup \cdots \sqcup B_K$ be a planted partition. Blocks $k \neq l$ are *adjacent* ($k \sim l$) if some edge joins them. For $i \in B_k$, let $e_i := |N(i) \setminus B_k|$ be the cross-degree, and let $e_{\max} := \max_i e_i$. Let $C_k \subset \mathbb{R}^p$ be convex sets with $\{z_i(0) : i \in B_k\} \subseteq C_k$; the natural choices are $C_k = \operatorname{conv}\{z_i(0) : i \in B_k\}$ and $C_k = \bar{B}(\mu_k, r_k)$. Put $D_k := \operatorname{diam} C_k$. Write $\bar{B}(\eta)$ for the closed ball of radius η at the origin, and $C + \bar{B}(\eta)$ for the Minkowski sum.

Theorem 2.1 (planted partition at the $\sqrt{c}$ scale)**.** *Let $t_0 \geq 2$, assume each induced subgraph $G[B_k]$ is connected, and let $\eta \geq 0$, $\delta > 1$ satisfy*

$$\text{(A1, mixing)} \qquad D_k + 2\eta \leq 1 \quad \text{for all } k, \tag{5}$$

$$\text{(A2, separation)} \qquad \operatorname{dist}(C_k, C_l) \geq \delta + 2\eta \quad \text{for all } k \sim l, \tag{6}$$

$$\text{(A3, drift budget)} \qquad e_{\max}\, \delta\, \frac{(t_0 - 1)^{1-\delta^2}}{\delta^2 - 1} \leq \eta. \tag{7}$$

Then:

(a) $z_i(t) \in C_k + \bar{B}(\eta)$ for all $t \geq 0$ and $i \in B_k$;

(b) each block converges to a single point: $z_i(t) \to z_k^\infty \in C_k + \bar{B}(\eta)$ for all $i \in B_k$;

(c) $\|z_k^\infty - z_l^\infty\| \geq \delta > 1$ for all $k \sim l$;

(d) the limiting partition, i.e. the connected components of the graph of edges of zero limiting length, is exactly $\{B_1, \ldots, B_K\}$.

Proof. Put $\varepsilon_t := e_{\max}\,\delta\, s_t^{-\delta^2}$ and $\eta_t := \sum_{u<t}\varepsilon_u$. By the integral comparison in Lemma 1.1(f) and (7), $\eta_t \le \eta$ for all t. We prove by induction the claim $H(t)$: $z_i(t) \in C_k + \bar{B}(\eta_t)$ for all k and $i \in B_k$. $H(0)$ holds.

Assume $H(t)$. *Cross distances.* If $i \in B_k$, $j \in B_l$ and $\{i,j\} \in E$ with $k \neq l$, then $k \sim l$ and $\|z_i(t) - z_j(t)\| \ge \operatorname{dist}(C_k, C_l) - 2\eta_t \ge \delta$ by (6). *Decomposition.* For $i \in B_k$ write $z_i(t+1) = y_i + \xi_i$ with

$$y_i := z_i + \sum_{j \in N(i)\cap B_k} \alpha_{ij}(z_j - z_i), \qquad \xi_i := \sum_{j \in N(i)\setminus B_k} \alpha_{ij}(z_j - z_i)$$

(all at time t). The point y_i is a convex combination of $\{z_j(t) : j \in B_k\}$: the coefficient of z_i is $\alpha_{ii} + \sum_{j\notin B_k}\alpha_{ij} \ge 0$, and the coefficients sum to 1. Hence $y_i \in C_k + \bar{B}(\eta_t)$, a convex set. For the cross term, $\pi_i \ge 1$ and $d_{ij} := \|z_i - z_j\| \ge \delta$ give

$$\|\xi_i\| \le \sum_{j \in N(i)\setminus B_k} d_{ij}\, e^{-d_{ij}^2 L_t} \le e_i\,\delta\, e^{-\delta^2 L_t} \le \varepsilon_t,$$

because $x \mapsto x e^{-x^2 L}$ is decreasing on $[1/\sqrt{2L}, \infty)$, and $\delta > 1 \ge 1/\sqrt{2\log 2}$. So $z_i(t+1) \in C_k + \bar{B}(\eta_t + \varepsilon_t)$, which is $H(t+1)$. This proves (a).

(b) For a within-block edge $\{i,j\} \subseteq B_k$, (a) and (5) give $\|z_i(t) - z_j(t)\| \le D_k + 2\eta \le 1$. Hence $w_{ij}(t) \ge s_t^{-1}$ and $\sum_t w_{ij}(t) = \infty$. By Proposition 1.5(ii) the endpoints share a limit. Since $G[B_k]$ is connected, all of B_k shares one limit z_k^∞, and it lies in the closed set $C_k + \bar{B}(\eta)$.

(c) $\|z_k^\infty - z_l^\infty\| \ge \operatorname{dist}(C_k, C_l) - 2\eta \ge \delta$.

(d) follows from (b) and (c). □

Corollary 2.2 (ball form)**.** *Let μ_k be any centres and $r_k := \max_{i\in B_k}\|z_i(0) - \mu_k\|$. The conclusions of Theorem 2.1 hold if, for some $\eta \ge 0$ and $\delta > 1$ satisfying* (7)*,*

$$r_k \le \tfrac{1}{2} - \eta \quad \text{and} \quad \|\mu_k - \mu_l\| \ge r_k + r_l + \delta + 2\eta \quad (k \sim l).$$

In physical units: $r \le \sqrt{c}\,(\frac{1}{2} - \eta)$ and $\Delta \ge 2r + (\delta + 2\eta)\sqrt{c}$. The old sufficient condition (`planted_certificate`) required $r \le \sqrt{c}/(4\sqrt{\rho})$ with ρ the hop radius of the blocks, and $\Delta \ge 4r + \dots$.

Proof. Take $C_k = \bar{B}(\mu_k, r_k)$, so $D_k = 2r_k$ and $\operatorname{dist}(C_k, C_l) \ge \|\mu_k - \mu_l\| - r_k - r_l$. □

Corollary 2.3 (the separation threshold is $\kappa(t_0)$ to leading order)**.** *Fix $e_{\max} \ge 1$ and $\theta \in (0,1]$, and let $\ell := \log(t_0 - 1) > e$. Define*

$$\Gamma(t_0) := \sqrt{1 + \tfrac{2\log\ell}{\ell}}\,\Big(1 + \frac{e_{\max}}{\ell\log\ell}\Big).$$

If $D_k \le 1 - \theta$ for all k, $e_{\max}\sqrt{1 + 2\log\ell/\ell} \le \theta\,\ell\log\ell$, and $\operatorname{dist}(C_k, C_l) \ge \Gamma(t_0)$ for all $k \sim l$, then the conclusions of Theorem 2.1 hold. Moreover $\Gamma(t_0) = 1 + (1 + o(1))\frac{\log\log t_0}{\log t_0}$.

Proof. Let $\gamma := 2\log\ell/\ell \in (0,1)$, $\delta := \sqrt{1+\gamma}$, and $\eta := e_{\max}\delta/(2\ell\log\ell)$. Then $(t_0 - 1)^{1-\delta^2} = e^{-\gamma\ell} = \ell^{-2}$, so (7) holds with equality. Condition (5) holds because $2\eta \le \theta$, and (6) holds because $\delta + 2\eta = \Gamma(t_0)$. Finally $\sqrt{1+\gamma} \le 1 + \log\ell/\ell$, and $\ell \sim \log t_0$. □

Remark 2.4 (sharpness of the separation term). By Theorem 1(v) of the main text, $\kappa(t_0; \rho_A, \rho_B) = 1 + (1 + o(1))\log\log t_0/\log t_0$ for every topology, so $\Gamma(t_0) - \kappa(t_0;\rho) = o(\log\log t_0/\log t_0)$. Conversely, for block-regular two-block graphs with collapsed blocks ($D_k = 0$), separation $\ge \kappa(t_0; \rho_A, \rho_B)$ is *necessary* by the two-block law. Hence (6) is sharp to leading order. The cut ratio ρ enters κ only at second order, while our condition uses $e_{\max}$ at second order; see Remark 2.7.

Proposition 2.5 (the mixing constant in (5) is sharp)**.** *For every $D > 1$ and $t_0 \geq 2$ there are a graph with a single block B such that $G[B]$ is connected, and features with* $\operatorname{diam}\{z_i(0)\} = D$*, for which the limit is* not *a consensus: the block splits into two clusters at distance > 1. Consequently no condition of the form $\max_k D_k \leq f(t_0)$ with $f(t_0) > 1$ (equivalently $r_k \leq f(t_0)/2$) can guarantee the planted partition, whatever the graph.*

Proof. Take two cliques Q_1, Q_2 of size k, joined by a perfect matching, with $z \equiv 0$ on Q_1 and $z \equiv D$ on Q_2 (scalars). Each node sees k nodes of its own clique (including itself) at distance 0 and one node at distance d. The features stay clique-constant, and the gap obeys $d' = d\,\frac{k-a}{k+a} \geq d(1-2a/k)$ with $a = s_t^{-d^2}$. Let $\delta := (1+D)/2$. If $d_u \geq \delta$ for all $u < t$, then $d_t \geq D\big(1 - \frac{2}{k}\,\frac{(t_0-1)^{1-\delta^2}}{\delta^2-1}\big) \geq \delta$, provided $k \geq \frac{2D(t_0-1)^{1-\delta^2}}{(\delta^2-1)(D-\delta)}$. By induction the gap stays $\geq \delta > 1$. □

Remark 2.6. Numerically, at the E4 offset $t_0 = 602$: with $k = 100$ the block of diameter $D = 1.1$ ends at gap 1.071; with $k = 1000$ even $D = 1.05$ persists (gap 1.044). This uses `core.two_block_final` with $\rho_A = \rho_B = 1/k$.

Remark 2.7 (where the two scales are, and what is not captured). The proof has two time scales. Within blocks, the weights are non-summable ($\geq s_t^{-1}$), and the blocks collapse over polynomially long times. Across blocks, the weights are summable ($\leq s_t^{-\delta^2}$), and the blocks drift by at most η *in total*. The invariant convex sets $C_k + \bar{B}(\eta)$ handle both at once, and the infinite-flow theorem (Lemma 1.4) replaces the path-product Dobrushin argument, which is what produced the factor $1/\sqrt{\rho}$. After a block collapses, the drift rate of node i improves from e_i to $\rho_i = e_i/|N[i] \cap B_k|$, because $\pi_i \to |N[i] \cap B_k|$. Exploiting this (replacing $e_{\max}$ by the cut ratio in (7)) needs an *explicit* within-block collapse time. Lemma 1.4 is qualitative, and quantitative versions for general graphs carry constants exponential in $|B_k|$. This second-order improvement is therefore not proved here.

2.1 Check against experiment E4

E4: $K = 4$ blocks of 150 nodes, SBM $p_{\text{in}} = 0.2$, $p_{\text{out}} \in \{0, \ldots, 0.05\}$, $p = 8$, means at mutual distance $\Delta = 3$, isotropic noise σ, $t_0 = n + 2 = 602$, $\kappa(602) \approx 1.30$, 25 runs per σ. We regenerate the exact initial conditions (`theory/check_E4.py`) and test (5)–(7). The hull form uses $C_k = \operatorname{conv}(\text{block } k)$, with dist lower-bounded by projection onto the direction joining the block means; the ball form is Corollary 2.2 with μ_k the block means. We scan $\eta \in (0, (1-D)/2]$ and take the smallest admissible δ.

σ	$\max_k D_k$ (runs)	r (runs)	old	ball	hull	survives
0.00	0	0	5	25	25	25
0.03	[0.228, 0.292]	[0.142, 0.184]	23	25	25	25
0.06	[0.464, 0.553]	[0.277, 0.335]	0	25	25	25
0.10	[0.788, 0.937]	[0.476, 0.578]	0	11	**25**	25
0.15	[1.155, 1.384]	[0.714, 0.846]	0	0	0	25
0.20	[1.601, 2.051]	[0.952, 1.268]	0	0	0	25
0.25	[1.925, 2.329]	[1.196, 1.410]	0	0	0	25
0.30	[2.401, 3.076]	[1.355, 1.901]	0	0	0	1

At $\sigma = 0.1$ the binding condition is (5) ($D \leq 0.937$); separation has a large margin (hull distance ≥ 2.35 against the required $\delta + 2\eta \leq 1.74$, with $e_{\max} \leq 40$). The condition is never violated: every certified run indeed keeps the planted partition. For $\sigma \geq 0.15$ *every* run has $\max_k D_k > 1$. By Proposition 2.5, no theorem that sees only block diameters, connectivity, degrees and separations

can certify these runs. Survival up to $\sigma = 0.25$ is an effect of dense blocks plus high-dimensional concentration (see the conjecture in `SUMMARY.md`).

3 Task B: the three-layer chain

Three layers with layer features $x_1 \le x_2 \le x_3$ and gaps $u = x_2 - x_1$, $v = x_3 - x_2$. Let $w_1 = s_t^{-u^2}$, $w_2 = s_t^{-v^2}$, $D_1 = 1 + w_1$, $D_2 = 1 + w_1 + w_2$, $D_3 = 1 + w_2$. Then

$$x_1' = \frac{x_1 + w_1 x_2}{D_1}, \quad x_2' = \frac{w_1 x_1 + x_2 + w_2 x_3}{D_2}, \quad x_3' = \frac{w_2 x_2 + x_3}{D_3},$$

which is (4) with $r = 3$. We write $D := u + v$, $h := u + v/2$ (distance from x_1 to the midpoint of x_2, x_3) and $\tilde{h} := v + u/2$. Gap 1 *persists* if $\liminf u > 0$ and *merges* if $u \to 0$. By Proposition 1.5 applied to $G_{3,m}$, the limits u_∞, v_∞ exist and lie in $\{0\} \cup [1, \infty)$.

Lemma 3.1 (identities)**.** *With F_t as in Lemma 1.1 (applied to u with $a = w_1$, or to v with $a = w_2$):*

$$u' = F_t(u) + \frac{w_2}{D_2}\Big(v + \frac{w_1 u}{D_1}\Big), \qquad v' = F_t(v) + \frac{w_1}{D_2}\Big(u + \frac{w_2 v}{D_3}\Big), \tag{8}$$

$$h' = h - w_1\Big[u\Big(\frac{1}{2D_2} + \frac{1}{D_1}\Big) + \frac{w_2 v}{2D_2 D_3}\Big], \qquad \tilde{h}' = \tilde{h} - w_2\Big[v\Big(\frac{1}{2D_2} + \frac{1}{D_3}\Big) + \frac{w_1 u}{2D_1 D_2}\Big], \tag{9}$$

$$D' = D - \frac{w_1 u}{D_1} - \frac{w_2 v}{D_3}. \tag{10}$$

Consequently $h, \tilde{h}, D$ are nonincreasing, $u \le h$, $v \le \tilde{h}$, $u, v \le D \le D_0$, and

$$(1 - 2w_1)u \;\le\; u' \;\le\; (1 - w_1)u + w_2(u + v), \qquad (1 - 2w_2)v \;\le\; v' \;\le\; (1 - w_2)v + w_1(u + v), \tag{11}$$

$$0 \le h - h' \le \tfrac{3}{2}\, w_1(u + v). \tag{12}$$

Proof. Put $x_1 = 0$, $x_2 = u$, $x_3 = u + v$. Then $x_1' = w_1 u/D_1$ and $x_2' = (u(1 + w_2) + w_2 v)/D_2$, so $u' = u\frac{(1+w_2)(1+w_1) - w_1 D_2}{D_1 D_2} + \frac{w_2 v}{D_2} = u\frac{1 + w_2 - w_1^2}{D_1 D_2} + \frac{w_2 v}{D_2}$. Since $\frac{1+w_2-w_1^2}{D_1 D_2} - \frac{1-w_1}{1+w_1} = \frac{w_1 w_2}{D_1 D_2}$, this is the first identity in (8); the second follows by the mirror symmetry $x \mapsto -x$, $1 \leftrightarrow 3$. With $\Delta x_1 = w_1 u/D_1$, $\Delta x_2 = (w_2 v - w_1 u)/D_2$ and $\Delta x_3 = -w_2 v/D_3$, we get $h' - h = \frac{\Delta x_2 + \Delta x_3}{2} - \Delta x_1$. Using $\frac{1}{D_2} - \frac{1}{D_3} = -\frac{w_1}{D_2 D_3}$ gives (9), and the mirror gives the $\tilde{h}$ identity. (10) is $\Delta x_3 - \Delta x_1$. The bounds use $1 - 2a \le \frac{1-a}{1+a} \le 1 - a$, $D_i \ge 1$, $w_i \le 1$, and, for (12), $\frac{1}{2D_2} \le \frac{1}{2}$, $\frac{1}{D_1} \le 1$, $\frac{w_2}{D_2 D_3} \le \frac{w_2}{(1+w_2)^2} \le \frac{1}{4}$. □

(These identities were checked numerically on $2 \cdot 10^5$ random points against `core.chain_step`; maximal error $2.7 \cdot 10^{-15}$. See `theory/check_identities.py`.)

3.1 An exact persistence criterion

Proposition 3.2 (exact criterion)**.** *Let $t_0 > 1$ and $u_0, v_0 \ge 0$.*

(a) If $u(t_1) + v(t_1)/2 < 1$ for some $t_1 \ge 0$, then gap 1 merges.

(b) If $u_0 + v_0 < 2$, then gap 1 persists $\iff$ $h(t) \ge 1$ for all t. In that case $u_\infty = h_\infty \ge 1$ and $v_\infty = 0$: layers $2, 3$ merge, and the merged block stays at distance h_∞ from layer 1.

(c) For every t_0: gap 1 merges whenever $u_0 + v_0/2 < 1$. Hence the rescue boundary obeys $u^*(v_0) \geq 1 - v_0/2$, i.e. $\geq \sqrt{c} - v_0/2$ in physical units.

(d) (symmetric line) If $u_0 = v_0 < 1$, then both gaps merge.

Proof. (a) $u_\infty \leq h_\infty \leq h(t_1) < 1$, so $u_\infty = 0$ by the dichotomy. (b) ($\Leftarrow$) If $u_\infty = 0$ then $h_\infty = v_\infty/2 \leq D_\infty/2 \leq D_0/2 < 1$, which is a contradiction; so $u_\infty \geq 1$. If moreover $v_\infty > 0$ then $v_\infty \geq 1$, so $D_\infty \geq 2 > D_0$, contradicting the monotonicity of D. Hence $v_\infty = 0$ and $u_\infty = h_\infty$. ($\Rightarrow$) The same argument gives $v_\infty = 0$, so $h(t) \geq h_\infty = u_\infty \geq 1$. (c) is (a) at $t_1 = 0$. (d) The dynamics commute with the mirror, so $u(t) = v(t)$ for all t. If gap 1 persisted, (b) would force $v_\infty = 0 \neq u_\infty$. □

3.2 Rescue for every t_0 and every v_0, in one step

Theorem 3.3 (one-step rescue)**.** *Let $t_0 \geq 2$, $L = \log t_0$, $v_0 > 0$. Put $w_* := t_0^{-v_0^2}$, $p_0 := \frac{w_* v_0}{1+w_*}$, $d_- := 1 - p_0$,*

$$p := \frac{w_* v_0}{1 + t_0^{-d_-^2} + w_*}, \qquad m := \max\{d_-^2, \tfrac{3}{2L}\}, \qquad M := 1 + (4mL - 2)_+ \, t_0^{-m}, \qquad R := p/M.$$

Then $R > 0$, and gap 1 persists for every *$u_0 \geq \kappa(t_0) - R$. In particular, for every $t_0 \geq 2$ and every $v_0 > 0$ the isolated rule "gap persists iff $u_0 \geq \kappa(t_0)$" is wrong on the nonempty interval $[\kappa(t_0) - R, \kappa(t_0))$: the neighbour pushes u across the critical curve in one step.*

Proof. First, $p_0 \leq \sup_v v e^{-v^2 L} = (2eL)^{-1/2} < 1$ for $L \geq \log 2$, so $d_- > 0$, and $R \leq p \leq p_0 < \kappa(t_0) - d_-$ because $\kappa > 1$. For $u_0 \geq \kappa(t_0)$ use Lemma 1.3. Now let $u_0 \in [\kappa - R, \kappa)$; then $u_0 > d_-$. By (8) at $t = 0$,

$$u(1) \;\geq\; F_0(u_0) + \frac{w_* v_0}{1 + t_0^{-u_0^2} + w_*} \;\geq\; F_0(u_0) + p.$$

By the mean value theorem, $F_0(\kappa) - F_0(u_0) \leq (\kappa - u_0) \sup_{d \geq d_-} F_0'(d)$, and $F_0'(d) - 1 = \frac{a(4d^2 L - 2 - 2a)}{(1+a)^2} \leq G(d^2)$ with $G(m') := (4m'L - 2)_+ e^{-m'L}$. G vanishes on $[0, \frac{1}{2L}]$, increases on $[\frac{1}{2L}, \frac{3}{2L}]$ and decreases afterwards, so $\sup_{m' \geq d_-^2} G(m') = G(m)$. Hence $F_0(\kappa) - F_0(u_0) \leq RM = p$, and $u(1) \geq F_0(\kappa(t_0)) = \kappa(t_0 + 1)$ by Lemma 1.1(d). Persistence follows from Lemma 1.3. □

Remark 3.4 (size of the one-step rescue). As $t_0 \to \infty$ with $v_0 = x/\sqrt{L}$: $p_0 \to 0$, $M \to 1$, and $\sup_{v_0} R = (c_1 + o(1))/\sqrt{\log t_0}$ with $c_1 = \max_x x e^{-x^2}/(1 + e^{-x^2}) = 0.2779$, attained at $x = 0.860$. Explicit values (`theory/constants.py`; κ_{hi} is the validated enclosure from E1, and κ_+ for $t_0 = 10^{10}$):

t_0	optimal v_0	$\max R$	$\max R/\kappa_{\mathrm{hi}}$	$0.278/\sqrt{\log t_0}$
5	0.649	0.0925	0.055	0.219
11	0.541	0.0839	0.055	0.179
100	0.398	0.0930	0.068	0.130
10^3	0.327	0.0976	0.075	0.106
10^6	0.231	0.0747	0.062	0.075
10^{10}	0.179	0.0579	0.051	0.058

The one-step rescue is qualitative (5–7.5% of κ). The macroscopic rescue (25–33%) needs $\gtrsim t_0^{v_0^2}$ steps, which is the content of Theorem 3.8.

3.3 The scaling limit of the rescue boundary

Throughout, $\kappa = \kappa(t_0)$, $u_0 = y\kappa$, $v_0 = x\kappa$, and

$$\varphi(x) := \max\{x,\ 1 - x/2\} \quad (0 < x < 1), \qquad \varphi(x) := 1 \quad (x > 1).$$

The three window lemmas below are stated in absolute units, for $t_0 \geq 2$ and an integer horizon $T \leq t_0$, so that $t_0 \leq s_t \leq 2t_0$ for $t \leq T$.

Lemma 3.5 (persistence window)**.** *Let $0 < b < a$, $\epsilon := T t_0^{-a^2}$ and $E := \exp(-T(2t_0)^{-b^2})$. If $u_0 \geq a + 2D_0\epsilon$ and $v_0 + D_0\epsilon \leq b$, then*

$$u(T) \ \geq \ u_0 + \tfrac{v_0}{2} - 2D_0\epsilon - \tfrac{v_0}{2}E.$$

Consequently gap 1 persists if $u_0 + \frac{v_0}{2} - 2D_0\epsilon - \frac{v_0}{2}E \geq \kappa(t_0)$.

Proof. By induction on $t \leq T$: $u(t) \geq u_0 - 2D_0 t\, t_0^{-a^2} \geq a$ and $v(t) \leq v_0 + D_0 t\, t_0^{-a^2} \leq b$. Indeed, if $u(t) \geq a$ then $w_1(t) \leq t_0^{-a^2}$, and (11) with $u, v \leq D_0$ gives $u(t+1) \geq u(t) - 2D_0 t_0^{-a^2}$ and $v(t+1) \leq v(t) + D_0 t_0^{-a^2}$. Then $w_2(t) \geq (2t_0)^{-b^2}$, so $v(t+1) \leq (1 - (2t_0)^{-b^2})v(t) + D_0 t_0^{-a^2}$ and $v(T) \leq v_0 E + D_0\epsilon$. By (12), $h(T) \geq h_0 - \frac{3}{2}D_0\epsilon$. Hence $u(T) = h(T) - v(T)/2$ satisfies the bound. Finally $\kappa(t_0 + T) \leq \kappa(t_0)$ (Lemma 1.1(d)), and Lemma 1.3 applies. □

Lemma 3.6 (race window)**.** *Let $0 < \bar{u} < v_{\mathrm{lo}}$, $\epsilon' := T t_0^{-v_{\mathrm{lo}}^2}$ and $E' := \exp(-T(2t_0)^{-\bar{u}^2})$. If $u_0 + D_0\epsilon' \leq \bar{u}$ and $v_0(1 - 2\epsilon') \geq v_{\mathrm{lo}}$, then $u(T) \leq u_0 E' + D_0\epsilon'$. Hence gap 1 merges if*

$$u_0 E' + D_0\epsilon' + \tfrac{v_0}{2} + \tfrac{u_0}{4} < 1.$$

Proof. By induction on $t \leq T$: $v(t) \geq v_0(1 - 2t_0^{-v_{\mathrm{lo}}^2})^t \geq v_0(1 - 2\epsilon') \geq v_{\mathrm{lo}}$, by the lower bound in (11) with $w_2 \leq t_0^{-v_{\mathrm{lo}}^2}$. Also $u(t) \leq u_0 + D_0 t\, t_0^{-v_{\mathrm{lo}}^2} \leq \bar{u}$, by the upper bound in (11) with $w_2(u+v) \leq D_0 t_0^{-v_{\mathrm{lo}}^2}$. Then $w_1(t) \geq (2t_0)^{-\bar{u}^2}$, so $u(t+1) \leq (1 - (2t_0)^{-\bar{u}^2})u(t) + D_0 t_0^{-v_{\mathrm{lo}}^2}$, which gives $u(T) \leq u_0 E' + D_0\epsilon'$. Moreover $v(T) \leq \tilde{h}(T) \leq \tilde{h}_0 = v_0 + u_0/2$. Hence $h(T) = u(T) + v(T)/2 < 1$, and Proposition 3.2(a) applies. □

Lemma 3.7 (persistent neighbour)**.** *Let $v_{\mathrm{lo}} > 1$ and $P_\infty := (t_0 - 1)^{1 - v_{\mathrm{lo}}^2}/(v_{\mathrm{lo}}^2 - 1)$. If $v_0(1 - 2P_\infty) \geq v_{\mathrm{lo}}$ and $u_0 + (v_{\mathrm{lo}} + 1)P_\infty < 1$, then gap 1 merges (and gap 2 persists).*

Proof. We show by induction on t that $v(t) \geq v_{\mathrm{lo}}$ and $u(t) \leq u_0 + (v_{\mathrm{lo}} + 1)\sum_{u' < t} s_{u'}^{-v_{\mathrm{lo}}^2} < 1$. First, $v(t) \geq v_0 \prod_{u' < t}(1 - 2s_{u'}^{-v_{\mathrm{lo}}^2}) \geq v_0(1 - 2P_\infty)$. Second, the push in (8) is at most $w_2 v + w_2 u \leq v_{\mathrm{lo}} s_t^{-v_{\mathrm{lo}}^2} + s_t^{-v_{\mathrm{lo}}^2}$, because $x \mapsto x s^{-x^2}$ is decreasing for $x \geq 1/\sqrt{2\log s}$ and $u < 1$; and $F_t(u) \leq u$. Hence $u_\infty \leq u_0 + (v_{\mathrm{lo}} + 1)P_\infty < 1$, so $u_\infty = 0$. □

Theorem 3.8 (scaling limit of the rescue boundary)**.** *Let $x \in (0,1) \cup (1,\infty)$, $y > 0$, $y \neq \varphi(x)$. There is an explicit $T^*(x,y)$ such that for all $t_0 \geq T^*$ the chain started from $(u_0, v_0) = (y\kappa(t_0), x\kappa(t_0))$ satisfies: gap 1 persists if $y > \varphi(x)$, and merges if $y < \varphi(x)$. Explicitly, with $\bar{\kappa} := \kappa_+(T^*)$ (Lemma 1.1(f); any certified upper bound for $\kappa(T^*)$ may be used), $T = \lceil t_0^\beta \rceil$, $e := 2t_0^{\beta - a^2}$ etc., the following conditions at $t_0 = T^*$ suffice. All of them get weaker as t_0 grows, because $\bar{\kappa}$ bounds $\kappa(t_0)$ for all $t_0 \geq T^*$ by Lemma 1.1(e).*

(i) *($x<1$, $\varphi(x)<y<1$; persistence) pick $x\bar\kappa<b<\sqrt{\beta}<a<y$, $\beta<1$, with $t_0^{\beta}+1\le t_0$ and*

$$e=2t_0^{\beta-a^2},\qquad y-2(x+1)\bar\kappa\,e\ge a,\qquad x\bar\kappa+(x+1)\bar\kappa\,e\le b,$$
$$y+\tfrac{x}{2}-1\ge 2(x+1)e+\tfrac{x}{2}\exp\bigl(-2^{-b^2}t_0^{\beta-b^2}\bigr).$$

(ii) *($y<1-x/2$, $x<2$; merge) $\bar\kappa\,(y+x/2)<1$.*

(iii) *($2/3<x<1$, $1-x/2\le y<x$; merge) pick $y\bar\kappa<\bar u<\sqrt{\beta}<v_{\rm lo}<x$, $t_0^{\beta}+1\le t_0$, $e'=2t_0^{\beta-v_{\rm lo}^2}$, and*

$$y\bar\kappa+(x+y)\bar\kappa\,e'\le\bar u,\qquad x(1-2e')\ge v_{\rm lo},$$
$$y\bar\kappa\,e^{-2^{-\bar u^2}t_0^{\beta-\bar u^2}}+(x+y)\bar\kappa\,e'+\bar\kappa\bigl(\tfrac{x}{2}+\tfrac{y}{4}\bigr)<1.$$

(iv) *($x>1$, $y<1$; merge) pick $1<v_{\rm lo}<x$ with $x(1-2P_\infty)\ge v_{\rm lo}$ and $y\bar\kappa+(v_{\rm lo}+1)P_\infty<1$.*

(v) *($y\ge 1$) persistence for every t_0 (Lemma 1.3).*

Each condition holds for all large t_0 because $\bar\kappa\to 1$ (Lemma 1.1(f)), $t_0^{\beta-a^2},t_0^{\beta-v_{\rm lo}^2},P_\infty\to 0$, and $t_0^{\beta-b^2},t_0^{\beta-\bar u^2}\to\infty$; and the regions (i)–(v) cover all (x,y) with $x\ne 1$, $y\ne\varphi(x)$.

Proof. (i) Apply Lemma 3.5 with these a,b,T, using $\kappa\ge 1$ (so $u_0\ge y$), $\kappa\le\bar\kappa$, $D_0=(x+y)\kappa\le(x+1)\bar\kappa$, $t_0^{\beta}\le T\le 2t_0^{\beta}$ (so $\epsilon\le e$ and $T(2t_0)^{-b^2}\ge 2^{-b^2}t_0^{\beta-b^2}$). The final inequality of Lemma 3.5, divided by κ, reads $y+x/2-1\ge 2(x+y)\epsilon+\frac{x}{2}E$. Such a,b exist since $y>x$, and the left side is positive since $y>1-x/2$. (ii) $u_0+v_0/2=\kappa(y+x/2)<1$; use Proposition 3.2(c). (iii) Lemma 3.6 with $D_0\le(x+y)\bar\kappa$, $v_0\ge x$, $u_0\le y\bar\kappa$, and $v_0/2+u_0/4\le\bar\kappa(x/2+y/4)$. As $t_0\to\infty$ the left side of the last condition tends to $x/2+y/4<3/4$. (iv) Lemma 3.7 with $v_0\ge x$, $u_0\le y\bar\kappa$. Coverage: for $x\le 2/3$, $\varphi=1-x/2$ and $y<\varphi$ is (ii); for $2/3<x<1$, $\varphi=x$ and $y<\varphi$ is (ii)∪(iii); for $x>1$, $\varphi=1$ and $y<1$ is (iv); and $y>\varphi$ is (i) or (v). □

Corollary 3.9 (normalised boundary; the minimum 2/3)**.** *Let $U_+(v_0;t_0):=\inf\{u:$ gap 1 persists from $(u',v_0)\ \forall u'\ge u\}$ and $U_-(v_0;t_0):=\sup\{u:$ gap 1 merges from $(u',v_0)\ \forall u'\le u\}$, so that $U_-\le U_+$. Then for every $x\in(0,1)\cup(1,\infty)$,*

$$\lim_{t_0\to\infty}\frac{U_\pm(x\kappa(t_0);t_0)}{\kappa(t_0)}=\varphi(x),\qquad\text{and}\qquad\lim_{t_0\to\infty}\inf_{v_0>0}\frac{U_\pm(v_0;t_0)}{\kappa(t_0)}=\frac{2}{3}=\varphi\bigl(\tfrac{2}{3}\bigr).$$

Proof. Pointwise: all conditions in Theorem 3.8 are monotone in y within each region (larger y only helps (i), smaller y only helps (ii)–(iv); in (i) use $x+1\ge x+y'$). So for each $\epsilon>0$, eventually gap 1 persists for all $y'\ge\varphi(x)+\epsilon$ and merges for all $y'\le\varphi(x)-\epsilon$. Minimum: $\limsup\inf_{v_0}U_+/\kappa\le\lim U_+(\frac{2}{3}\kappa)/\kappa=\frac{2}{3}$. For the lower bound, fix $\epsilon\in(0,\frac{1}{6})$ and $y\le\frac{2}{3}-\epsilon$; we show merging uniformly in $x>0$ for large t_0. If $x\le\frac{2}{3}+\epsilon$, then $y+x/2\le 1-\epsilon/2$, and (ii) holds once $\kappa_+(t_0)<1/(1-\epsilon/2)$. If $\frac{2}{3}+\epsilon\le x\le\frac{4}{3}$, apply Lemma 3.6 with $\bar u=\frac{2}{3}-\frac{\epsilon}{2}$, $v_{\rm lo}=\frac{2}{3}+\frac{\epsilon}{2}$, $\beta=(\bar u^2+v_{\rm lo}^2)/2$, $D_0\le 2\kappa$. The last condition becomes $\bar\kappa(\frac{x}{2}+\frac{y}{4})\le\frac{5}{6}\bar\kappa<1$ up to vanishing terms, uniformly in x. If $x\ge\frac{4}{3}$, apply Lemma 3.7 with $v_{\rm lo}=\frac{7}{6}$, uniformly in x. Hence $U_-\ge(\frac{2}{3}-\epsilon)\kappa$ for all v_0 once t_0 is large. □

Remark 3.10 (comparison with the data and with E2b). The finite-depth minima $\min_{v_0}u^*/\kappa=0.770,0.751,0.722,0.695,0.682$ at $t_0=5,11,50,10^3,10^5$ (`results/theory_E2.json`, key `E2b`) decrease towards 2/3, and the observed argmin $v_0/\kappa\approx$ 0.62–0.64 is close to

2/3. A sharper finite-t_0 check (`theory/check_scaling.py`: merge iff $h < 1$, persist iff $u \geq$ the critical trajectory from κ_{hi}) at $t_0 = 10^5$ gives, for $x = 0.1, 0.3, 0.5, 0.6, \frac{2}{3}, 0.7, 0.8, 0.9$: $u^*/\kappa = 0.937, 0.837, 0.737, 0.692, 0.684, 0.703, 0.800, 0.900$, against $\varphi = 0.950, 0.850, 0.750, 0.700, 0.667, 0.700, 0.800, 0.900$. On the branch $\varphi = x$ the agreement is already exact at finite t_0, consistent with the symmetric line, Proposition 3.2(d). On the branch $1 - x/2$ the finite-t_0 boundary lies slightly *below* φ. A plausible (unproved) explanation: once $\{2, 3\}$ has merged, h loses mass at rate $\approx \frac{5}{4} w_1 u$ by (9), against $2w_1 u$ for the isolated pair. So the relevant finite-t_0 threshold for h is somewhat below $\kappa(t_0)$, while both tend to 1.

Remark 3.11 (how explicit is T^*). The constants in Theorem 3.8 are explicit but crude; `theory/constants.py` optimises (a, b, β) on a grid. Uniform persistence for $x \in [0.3, 0.5]$, $y \geq 0.9$ (a 10% rescue) holds for $t_0 \geq 2.0 \cdot 10^8$, and for $x \in [0.4, 0.6]$, $y \geq 0.85$ for $t_0 \geq 8.4 \cdot 10^{15}$. Near the corner $x = y = 2/3$, T^* is astronomically large ($5 \cdot 10^{110}$ for $x \in [0.55, 0.75]$, $y \geq 0.8$), because the proof needs a polynomial separation of the time scales $t_0^{v_0^2} \ll t_0^{u_0^2}$. Uniform merging in the race region holds for $x \in [0.85, 0.95]$, $y \leq 0.5$ from $t_0 \geq 1.9 \cdot 10^{10}$. For moderate t_0 the macroscopic rescue is established only by the validated computations (Theorem 3 of the main text, E2).

4 Summary of what is proved

- Proposition 1.5: on every graph, limits exist and every limiting edge length is 0 or $\geq \sqrt{c}$. This relies on Touri–Nedić [1].
- Theorem 2.1 and Corollaries 2.2–2.3: planted partition under $D_k \leq (1 - 2\eta)\sqrt{c}$ and $\mathrm{dist}(C_k, C_l) \geq (\delta + 2\eta)\sqrt{c}$. The separation threshold is $\kappa(t_0)$ to leading order, and the mixing constant is sharp (Proposition 2.5). E4: certified for $\sigma \leq 0.1$ (100/100 runs).
- Proposition 3.2: the exact criterion *gap 1 persists iff* $\inf_t(u + v/2) \geq 1$, the exact merge region $u_0 + v_0/2 < 1$ for every t_0, and the symmetric line.
- Theorem 3.3: one-step rescue for every $t_0 \geq 2$ and every $v_0 > 0$, with explicit depth.
- Theorem 3.8 and Corollary 3.9: the scaling limit $\varphi(x) = \max\{x, 1 - x/2\}$ $(x < 1)$, $\varphi = 1$ $(x > 1)$; $\min \varphi = 2/3$ at $x = 2/3$.

References

[1] B. Touri and A. Nedić. Product of random stochastic matrices. *IEEE Trans. Automatic Control* 59(2):437–448, 2014. arXiv:1110.1751.

[2] J. M. Hendrickx and J. N. Tsitsiklis. Convergence of type-symmetric and cut-balanced consensus seeking systems. *IEEE Trans. Automatic Control* 58(1):214–218, 2013. arXiv:1102.2361.